

Quadratic functions

A quadratic function has a nonzero squared-term coefficient.

$$f(x) = ax^2 + bx + c, \quad a \neq 0$$

What feature distinguishes its graph from a line?

The vertex

The vertex is a parabola's turning point.

$$f(x) = (x - 3)^2 + 2$$

Where is the vertex?

Opening direction

$$f(x) = -2(x + 1)^2 + 5$$

Does the parabola open upward or downward?

Axis of symmetry

$$f(x) = 3(x - 4)^2 - 7$$

What vertical line is the axis of symmetry?

Comparing widths

$$f(x) = x^2, \quad g(x) = 4x^2$$

Which parabola is narrower?

Transforming a parabola

$$f(x) = -\frac{1}{2}(x - 2)^2 + 6$$

How is its graph positioned relative to $y = x^2$?

Vertex from general form

$$h = -\frac{b}{2a}, \quad k = f(h)$$

What is the vertex of $f(x) = 2x^2 - 8x + 3$?

Completing the square

$$f(x) = x^2 + 6x + 5$$

What equivalent vertex form describes this function?

A vertical intercept

$$f(x) = -x^2 + 4x + 5$$

Where does the graph intersect the vertical axis?

Horizontal intercepts

$$f(x) = x^2 - 5x + 6$$

Where does the graph intersect the horizontal axis?

One horizontal intercept

$$f(x) = (x + 2)^2$$

How many distinct real zeros does this function have?

No real zeros

$$f(x) = 2(x - 1)^2 + 3$$

Why does this graph have no horizontal intercepts?

The discriminant and the graph

$$f(x) = x^2 + 2x + 5$$

What does the discriminant tell us about horizontal intercepts?

Domain and range

$$f(x) = -3(x - 2)^2 + 12$$

What are the domain and range?

Increasing and decreasing

$$f(x) = (x + 4)^2 - 1$$

On what intervals is the function decreasing and increasing?

The minimum value

$$f(x) = 2x^2 - 12x + 11$$

What is the minimum value, and at what input does it occur?

Building from a vertex

A parabola has vertex $(1, -4)$ and passes through $(3, 4)$.

What quadratic function has this graph?

Building from zeros

A quadratic has zeros -2 and 5 , and its graph passes through $(0, -20)$.

What quadratic function satisfies these conditions?

A maximum height

A model gives height in meters t seconds after launch.

$$h(t) = -5t^2 + 20t + 1$$

What maximum height does the model predict?

An area model

A rectangular enclosure uses 40 meters of fencing on all four sides. Let x be one side length.

What quadratic function gives its area, and what is its physical domain?

Maximizing area

For a rectangular enclosure, x is a side length in meters.

$$A(x) = x(20 - x), \quad 0 < x < 20$$

What dimensions maximize the area?

A line and a parabola

$$f(x) = x^2, \quad g(x) = x + 2$$

Where do the two graphs intersect?

The sign of a quadratic

$$f(x) = (x - 1)(x - 4)$$

For which inputs is the graph below the horizontal axis?